



LOST BOYS CYBER

THREAT INTELLIGENCE • MALWARE ANALYSIS • DETECTION ENGINEERING



FAA and TSA Aviation Cybersecurity Shortfalls Daily Sector Review

Threat Report

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AI Generation: This report was generated with AI assistance and is provided for informational purposes only. All findings, IOCs, detection rules, hunting queries, attribution assessments, and recommendations must be independently reviewed and validated by a qualified security professional before operational use.

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FAA and TSA Aviation Cybersecurity Shortfalls Daily Sector Review

1. Executive Summary

GAO-26-107693 found FAA and TSA collaborate on aviation cybersecurity but must address role, strategy, budget-reporting, and oversight shortfalls. Aviation operators should use the report as a governance and dependency-risk checklist.

This daily report is intended for defender prioritization rather than attribution finality. Source expansion beyond 24-72 hours was used where the best primary source remained recently relevant but was published earlier than today's run.

2. Intelligence Requirements

Question	Current Answer	Confidence
What changed that defenders should review today?	The risk is governance and control coverage rather than a single exploit: unclear responsibility boundaries and incomplete reporting can delay risk reduction across NAS, airport, airline, and passenger-processing dependencies.	High
Which environments should care most?	airports, airlines, NAS support organizations, aviation regulators.	Medium
What should the SOC do first?	Validate exposure, run the included telemetry hunts, and confirm response owners for likely impact paths.	High

3. Source Review & Confidence

Source	Contribution	Confidence
GAO report page	Primary GAO report landing page.	High
GAO PDF	Full report.	Medium
Aviation Today summary	Sector-oriented summary.	Medium
Cabrillo Club analysis	Secondary analysis of governance findings.	Medium

4. Threat Activity Overview

The risk is governance and control coverage rather than a single exploit: unclear responsibility boundaries and incomplete reporting can delay risk reduction across NAS, airport, airline, and passenger-processing dependencies.

The most important operational takeaway is not a single indicator; it is the behavior pattern and exposed dependency class. Defenders should use the detections and hunts below as starting points, then tune to local log schemas.

5. Affected Sectors and Exposure

Exposure Pattern	Why It Matters
Directly exposed technology or service	Increases likelihood of opportunistic targeting and automated probing.
Identity, email, remote-access, or collaboration dependency	Increases blast radius because successful access can bypass perimeter assumptions.

Supplier-managed or shared service	Creates concentration risk and incident-notification delays.
Operationally critical workflow	Converts cyber compromise into business or mission disruption.

6. Aviation and Aerospace Sector Impact Analysis

Sector Impact Field	Assessment
Affected segment	airports, airlines, NAS support organizations, aviation regulators
Operational relevance	The issue can disrupt passenger processing, mission systems, satellite/space dependencies, regulatory oversight, or aviation operational continuity depending on local exposure.
Likely exposure paths	Shared IT providers, remote access, SaaS/identity platforms, supplier connectivity, operational technology, ground-segment systems, and connected avionics support environments.
Third-party / supply-chain considerations	Aviation and aerospace operators should require incident notification, continuity procedures, and cyber-control evidence from shared technology providers and critical suppliers.
Defender actions	Map dependencies, confirm backup/manual procedures, monitor identity and remote-access telemetry, and rehearse escalation between SOC, operations control, safety, and vendor-management teams.

7. Tactics, Techniques, and Procedures

Tactic / Phase	Observed Technique	Evidence
Initial Access	Phishing, exposed service, supply-chain, or remote-access abuse	Reflected in the cited reporting and sector exposure.
Execution / Persistence	Scripted tooling, malware, credential/session abuse, or provider-side compromise	Varies by campaign; validate with endpoint and application telemetry.
Defense Evasion	Use of legitimate tools, trusted platforms, or edge infrastructure	Common pattern across the reviewed reporting.
Impact	Data theft, service disruption, ransomware/extortion, or operational degradation	Primary defender concern for airports, airlines, NAS support organizations, aviation regulators.

8. Infrastructure and Indicators

Indicator / Infrastructure	Type	Operational Use
No normalized hard IOC set in this automated review	Source limitation	Use behavior and exposure hunts; ingest vendor IOC bundles when available.
Rare external authentication sources	Behavioral indicator	Triage abnormal access into affected systems.

New or unusual outbound connections from edge/server workloads	Behavioral indicator	Hunt possible relay, malware, or exfiltration staging.
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9. Detection Engineering

9.1. Detection Summary

Signal	Telemetry Source	Analytic Goal
Rare admin or service login after advisory/report publication	Identity/app logs	Identify initial access or supplier compromise.
Unusual child processes from browsers, package managers, admin consoles, or service accounts	Endpoint telemetry	Catch execution following social engineering or supply-chain compromise.
New outbound connections to rare hosts from servers or build systems	Network/EDR	Surface C2, relay-node, or exfiltration behavior.

9.2. Sigma / Detection Content

```

title: Daily Threat Review Suspicious Post Access Behavior
status: experimental
description: Behavioral seed analytic for daily Lost Boys Cyber threat reports; tune service names and parent processes locally.
logsource:
  product: windows
  category: process_creation
detection:
  selection_parent:
    ParentImage|endswith:
      - '\chrome.exe'
      - '\msedge.exe'
      - '\node.exe'
      - '\npm.cmd'
      - '\powershell.exe'
  selection_child:
    Image|endswith:
      - '\powershell.exe'
      - '\cmd.exe'
      - '\wscript.exe'
      - '\rundll32.exe'
      - '\curl.exe'
  condition: selection_parent and selection_child
level: medium

```

9.3. Hunt Query

```

DeviceProcessEvents
| where Timestamp > ago(14d)
| where InitiatingProcessFileName in~ ("chrome.exe", "msedge.exe", "node.exe", "npm.cmd", "powershell.exe")
| where FileName in~ ("powershell.exe", "cmd.exe", "wscript.exe", "rundll32.exe", "curl.exe", "msiexec.exe")
| summarize events=count(), firstSeen=min(Timestamp), lastSeen=max(Timestamp) by DeviceName, InitiatingProcessFileName, FileName, ProcessCommandLine
| order by events desc

```

10. Threat Hunting

10.1. Hunt Hypothesis

If this activity is present in the environment, defenders will observe a combination of anomalous authentication, unusual process lineage, rare network destinations, and supplier or operational workflow disruptions.

10.2. Hunt Query

<pre>DeviceNetworkEvents where Timestamp > ago(14d) where InitiatingProcessFileName in~ ("powershell.exe", "cmd.exe", "node.exe", "npm.exe", "chrome.exe", "msedge.exe") summarize connections=count(), firstSeen=min(Timestamp), lastSeen=max(Timestamp), remotePorts=make_set(RemotePort, 10) by DeviceName, InitiatingProcessFileName, RemoteUrl, RemoteIP where connections > 3 order by connections desc</pre>
<pre>SigninLogs where TimeGenerated > ago(14d) summarize attempts=count(), failures=countif(ResultType != 0), countries=make_set(Location, 10) by UserPrincipalName, AppDisplayName, IPAddress where failures > 10 or array_length(countries) > 2</pre>

11. Risk Assessment

Risk Driver	Assessment
Likelihood	Medium for organizations matching the exposed technology/sector pattern.
Impact	Ranges from credential theft and data exposure to operational disruption or ransomware.
Confidence	High; local validation required before incident declaration.
Priority Environments	airports, airlines, NAS support organizations, aviation regulators.

12. Recommended Actions

Priority	Owner	Action
Critical	SOC	Run the included hunts and tune for named platforms, suppliers, or services in scope.
High	IT / Platform owners	Confirm exposed assets, administrative access paths, and recent patch/configuration posture.
High	Vendor management	Identify third-party dependencies and request incident/control attestations where relevant.
Medium	IR lead	Prepare containment and communications playbooks for the most likely impact path.

13. References

- [1] GAO report page: <https://www.gao.gov/products/gao-26-107693>
- [2] GAO PDF: <https://www.gao.gov/assets/gao-26-107693.pdf>
- [3] Aviation Today summary: <https://aviationtoday.in/united-states/gao-warns-faa-cybersecurity-gaps-could-leave-us-aviation-vulnerable-to-threats/>
- [4] Cabrillo Club analysis: <https://cabrilloclub.com/insights/war-room-aviation-cybersecurity-faa-and-tsa-are-collaborating-on-cybersecurity-but-need-2026-07-17>